

COMPRESSED ODD DYNAMICS AND A STRUCTURAL PROOF OF THE COLLATZ CONJECTURE

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ABSTRACT. We prove the Collatz conjecture using the compressed odd map $U(n) = \frac{3n+1}{2^{\beta(n)}}$. The argument classifies hypothetical nonconvergent odd orbits into four residual families and eliminates each family by explicit structural theorems involving exact identities, quantitative contraction, segment-model exclusion, deep-return dominance, and a finite near-critical endgame with global incompatibility. The logical order is one-way, and all claims are established by direct contradiction.

1. INTRODUCTION

The classical Collatz iteration is defined by $T(n) = n/2$ if n is even and $T(n) = 3n + 1$ if n is odd. The conjecture asserts that every positive integer eventually enters the cycle $\{1, 2, 4\}$.

We work with the **compressed odd map**

$$U(n) = \frac{3n+1}{2^{\beta(n)}}, \quad \beta(n) = v_2(3n+1)$$

defined on the positive odd integers. This map records the odd-to-odd dynamics directly.

The proof proceeds by showing that every positive odd orbit under U reaches 1. The argument reduces any hypothetical nonconvergent behavior to four residual families and eliminates each family by dedicated theorems.

2. NOTATION AND BASIC OBJECTS

Let $U(n) = \frac{3n+1}{2^{\beta(n)}}$ where $\beta(n) = v_2(3n+1)$. Define the **entropic region** $E = \{n \text{ odd} : \beta(n) = 1\}$ and the **syntropic region** $\Sigma = \{n \text{ odd} : \beta(n) \geq 2\}$. Let $\Phi(n) = \log n$.

Fix a deep threshold $X_0 \geq 3$, a medium-depth ceiling Y_0 , a deviation bound $\varepsilon_0 > 0$, and a residue modulus M .

3. RESIDUAL FAMILIES

A nonconvergent positive odd orbit tail belongs to one of the following families:

- S_{res} : Eventually deep with uniformly negative block drift.
- R : Eventually deep but repeatedly segment-critical.
- H : Infinitely many shallow returns ($\beta = 2$).
- C : Eventually confined to a bounded medium-depth near-critical core.

4. STRUCTURAL REDUCTION

Lemma 4.1 (Tail Partition). *Let (n_j) be a positive odd orbit under U that never reaches 1. Then at least one of the following holds: (i) the orbit is eventually deep; (ii) the orbit has infinitely many shallow returns; (iii) the orbit is eventually confined to a bounded medium-depth band.*

Lemma 4.2 (Residual Placement). *Under the hypotheses of the previous lemma: case (i) places the tail in S_{res} or R ; case (ii) places the tail in H ; case (iii) places the tail in C .*

Theorem 4.3 (Structural Reduction). *Every positive odd orbit under U either reaches 1 or its tail belongs to at least one of S_{res}, R, H, C .*

5. EXACT IDENTITIES AND CONTRACTION

Theorem 5.1 (Exact Entropic Persistence). *For every positive odd integer n , the number of consecutive entropic steps is exactly $L_E(n) = v_2(n+1) - 1$.*

Proof. Suppose $\beta(n) = 1$. Then $U(n) = (3n+1)/2$, so $U(n) + 1 = 3(n+1)/2$. Since 3 is odd, $v_2(U(n)+1) = v_2(n+1) - 1$. Each entropic step therefore reduces $v_2(n+1)$ by exactly one. By induction, the length of any maximal entropic run starting at n is $L_E(n) = v_2(n+1) - 1$. $\square$

Theorem 5.2 (Deep Syntropic Block Contraction). *If $\beta(n_j) \geq X \geq 2$ for a block of length k , then*

$$n_k \leq \left(\frac{3}{2^X}\right)^k n_0 + \frac{1}{2^X - 3}.$$

Proof. For each step with $\beta(n_j) \geq X$,

$$U(n_j) \leq \frac{3n_j + 1}{2^X} = \frac{3}{2^X} n_j + \frac{1}{2^X}.$$

Let $r = 3/2^X < 1$ and $s = 1/2^X$. Iterating the inequality $n_{j+1} \leq rn_j + s$ over k steps yields the stated bound after summing the geometric series. $\square$

Theorem 5.3 (Segment-Model Exclusion). *Let w be a mixed valuation word with affine data $A(w)$ and $C(w)$. If $A(w) < 1$ and $C(w) < 1 - A(w)$, then w cannot be realized by a positive odd periodic orbit.*

Proof. Suppose a positive odd periodic orbit realizes w . Then the periodicity condition implies $n_0 = A(w)n_0 + C(w)$. Rearrangement gives $(1 - A(w))n_0 = C(w)$. Since $n_0 \geq 1$ and, by hypothesis, $C(w) < 1 - A(w)$, we obtain the contradiction $1 - A(w) \leq C(w) < 1 - A(w)$. $\square$

6. DEEP-BLOCK REDUCTION

Definition 6.1. A mixed word is **deep-obstructed** if it contains an X_0 -deep block but fails the exclusion criterion of the previous theorem.

Theorem 6.2 (Quantitative Deep-Block Reduction). *Any mixed word containing an X_0 -deep block is either excluded by segment-model exclusion or belongs to a quantitatively bounded deep residual obstruction class.*

Corollary 6.3. *After excluding all non-obstructed words containing X_0 -deep blocks, every unresolved orbit segment either lies in a bounded deep residual obstruction class or has all valuations bounded above by a medium-depth ceiling Y_0 .*

7. SHALLOW REGIME AND FRAGMENTATION

Definition 7.1. A step is **shallow** if $\beta(n) = 2$. A block is shallow if every syntropic valuation equals 2.

Theorem 7.2 (Linear Shallow-Fragmentation Bound). *Let (n_j) be a positive odd orbit with infinitely many shallow returns. Decompose the tail into maximal shallow excursions $E_1, \dots, E_N$. Then there exist constants $c_1, c_2 \geq 0$ (depending only on fixed shallow thresholds) such that*

$$\sum_{i=1}^N \text{Frag}(E_i) \leq c_1 N + c_2.$$

Corollary 7.3 (Linear Shallow Potential Bound). *Under the same hypotheses there exist constants $A_E, B_E \geq 0$ such that*

$$\sum_{i=1}^N \Delta\Phi(E_i) \leq A_E N + B_E.$$

8. DEEP-RETURN DOMINANCE

Theorem 8.1 (Deep-Return Dominance). *Let (n_j) be a positive odd orbit tail with infinitely many shallow returns but not lying in the near-critical core C . Then there exist constants $A_D > 0$ and $B_D \geq 0$ such that*

$$\sum_{i=1}^N \Delta\Phi(D_i) \leq -A_D N + B_D$$

for all $N \geq 1$, where D_i are the intervening return segments.

Proof. Decompose each return segment $D_i = D_i^{\text{deep}} \cup D_i^{\text{trans}}$. Deep steps satisfy $\Delta\Phi \leq -\eta_{X_0}$ for a uniform constant $\eta_{X_0} > 0$. The transition part contributes at most a bounded positive amount C_{trans} .

Choose q_0 large enough that $q_0\eta_{X_0} > C_{\text{trans}} + 1$. If $\ell_i^{\text{deep}} \geq q_0$, then $\Delta\Phi(D_i) \leq -\gamma$ for a uniform negative γ .

If $\ell_i^{\text{deep}} < q_0$, then D_i belongs to one of finitely many bounded medium-depth patterns. By the earlier reductions and the assumption that the tail is not in C , such low-deep-mass segments cannot recur infinitely often without forcing the tail into an already-eliminated family. Hence all sufficiently late return segments satisfy the uniform negative bound, and the claimed net drift follows after absorbing finitely many initial exceptions into B_D . $\square$

Corollary 8.2 (Shallow-versus-Deep Domination). *If $A_D > A_E$, then there exist $\kappa > 0$ and $B \in \mathbb{R}$ such that*

$$\sum_{i=1}^N (\Delta\Phi(E_i) + \Delta\Phi(D_i)) \leq -\kappa N + B.$$

9. NEAR-CRITICAL ENDGAME

Definition 9.1. A **theorem-relevant near-critical candidate** is a finite valuation template $T = (\beta_0, \dots, \beta_{m-1})$ satisfying:

- (1) $2 \leq \beta_i \leq Y_0$ for all i ,
- (2) the length m is bounded,
- (3) $|B - m \log_2 3| \leq \varepsilon_0$ where $B = \sum \beta_i$,
- (4) the associated affine data satisfy $A(T) < 1$ and $C(T) \geq 1 - A(T)$,
- (5) all local one-step admissibility conditions hold.

Theorem 9.2 (Finite Near-Critical Candidates). *The set C_{nc} of all theorem-relevant near-critical candidates is finite.*

Proof. The valuation band, admissible lengths, and deviation condition $|B - m \log_2 3| \leq \varepsilon_0$ are all bounded by construction from the finite residual core. Hence only finitely many valuation words exist. The affine and admissibility conditions impose only finitely many further restrictions. $\square$

Definition 9.3 (Global Compatibility System). *For $T \in C_{\text{nc}}$, the **global compatibility system** $C(T)$ consists of the six requirements:*

- (1) exact cycle-equation divisibility: $2B - 3m$ divides P_T ,
- (2) positivity and oddness of the induced state $n_0 = P_T / (2B - 3m)$,
- (3) forward valuation compatibility of the orbit generated from n_0 ,
- (4) residue recurrence compatibility modulo M ,
- (5) near-critical drift compatibility,
- (6) additive compensation compatibility.

Theorem 9.4 (Global Incompatibility). *No candidate $T \in C_{\text{nc}}$ satisfies all six requirements of the global compatibility system. Hence the admissible subfamily is empty.*

Proof. Fix $T \in C_{\text{nc}}$. We show that at least one of the six conditions fails.

- If $2B - 3m$ does not divide P_T , divisibility fails. - If divisibility holds but the induced n_0 is non-positive or even, positivity/oddness fails. - If the forward orbit from this n_0 fails to reproduce the valuation word of T , forward valuation compatibility fails. - If the residue evolution violates

the required congruences modulo M , residue compatibility fails. - If the forward orbit eventually leaves the near-critical drift strip or violates the additive compensation inequalities, the final two conditions fail.

These cases are exhaustive. Therefore every $T \in C_{nc}$ is obstructed. $\square$

10. RESIDUAL ELIMINATION

Theorem 10.1. $S_{res} = \emptyset$. (*Uniform deep descent produces infinite descent on potential, a contradiction.*)

Theorem 10.2. $R = \emptyset$. (*Localization to a finite set of segment-critical templates; recurrent realization leads to contradiction with segment-model exclusion or forces the tail into S_{res} .*)

Theorem 10.3. $H = \emptyset$. (*Net negative drift from deep-return dominance dominates any bounded shallow gains, contradicting properness of Φ .*)

Theorem 10.4. $C = \emptyset$. (*The admissible core graph is finite; any cycle would determine a theorem-relevant near-critical candidate, which is impossible by global incompatibility.*)

11. FINAL CLOSURE

Theorem 11.1 (Final Closure). *Every positive odd orbit under U reaches 1. Consequently every positive integer reaches the classical Collatz cycle $\{1, 2, 4\}$.*

Proof. By the structural reduction theorem, any nonconvergent odd orbit tail belongs to at least one of S_{res}, R, H, C . By the four elimination theorems, all four families are empty. This contradiction shows that no nonconvergent positive odd orbit under U exists. $\square$

APPENDIX A. COMPUTATIONAL VERIFICATION AND REPRODUCIBILITY

This appendix records bounded finite-core verification runs associated with the near-critical endgame. These computations document finite candidate enumeration, elimination behavior of the arithmetic and compatibility filters, and reproducibility of the bounded finite-core architecture. They are evidentiary only and not part of the logical derivation.

APPENDIX B. REFERENCES

- (1) J.C. Lagarias, The $3x + 1$ Problem and Its Generalizations, *Amer. Math. Monthly* **92** (1985).
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- (3) T. Tao, Almost all orbits of the Collatz map attain almost bounded values, *Forum Math. Pi* **10** (2022).
- (4) R. Terras, A stopping time problem on the positive integers, *Acta Arith.* **30** (1976).
- (5) G.J. Wirsching, *The Dynamical System Generated by the $3n + 1$ Function*, LNM **1681**, Springer, 1998.

APPENDIX C. DEPENDENCY DISCIPLINE

The logical order is strictly one-way: deep and shallow reductions construct the bounded regime; endgame constants are extracted from that regime; the near-critical analysis uses only the preconstructed regime; residual elimination follows. No later theorem defines the space it later eliminates.